AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	22.4 / 3.69302
Time	2026-06-03 22:00 UTC
Location	29.5205, -95.3925
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	7 / 489	50.67 to 100; mean 84.1782	84.61 at 2026-06-03 21:55Z; dt 5 min
asos	temperature	degC	7 / 489	22 to 33.3333; mean 26.025	25 at 2026-06-03 21:55Z; dt 5 min
asos	wind_direction	degree	7 / 488	0 to 360; mean 91.0656	50 at 2026-06-03 21:55Z; dt 5 min
asos	wind_speed	m/s	7 / 489	0 to 9.77444; mean 3.1529	3.08666 at 2026-06-03 21:55Z; dt 5 min
noaa_gfs	gh_500	m	8 / 96	5867.27 to 5894.87; mean 5877.92	5868.27 at 2026-06-04 00:00Z; dt 120 min
noaa_gfs	pbl_height	m	8 / 96	14.9006 to 1854.26; mean 508.689	323.528 at 2026-06-04 00:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	8 / 96	46.2383 to 56.7462; mean 51.4141	51.2229 at 2026-06-04 00:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	8 / 96	1007.93 to 1015.86; mean 1012.41	1012 at 2026-06-04 00:00Z; dt 120 min
noaa_gfs	t_850	degC	8 / 96	16.22 to 19.6344; mean 17.2148	16.9288 at 2026-06-04 00:00Z; dt 120 min
noaa_gfs	u_10m	m/s	8 / 96	-5.08847 to 0.197017; mean -2.4204	-2.97816 at 2026-06-04 00:00Z; dt 120 min
noaa_gfs	v_10m	m/s	8 / 96	-3.20242 to 3.28825; mean 0.0527	0.127957 at 2026-06-04 00:00Z; dt 120 min
openaq	ozone	ppm	12 / 831	-0.002 to 0.058; mean 0.0185	0.014 at 2026-06-03 22:00Z; dt 0 min
openaq	pm10	ug/m3	2 / 145	9 to 78; mean 20.1241	11 at 2026-06-03 22:00Z; dt 0 min
openaq	pm25	ug/m3	26 / 2766	-4 to 42.5; mean 6.0708	7.4 at 2026-06-03 22:00Z; dt 0 min
openweather	cloud_cover	percent	4 / 288	0 to 100; mean 69.6979	95 at 2026-06-03 21:57Z; dt 2.4 min
openweather	humidity	percent	4 / 288	60 to 97; mean 84.6771	89 at 2026-06-03 21:57Z; dt 2.4 min
openweather	precipitation	mm/h	4 / 288	0 to 9.99; mean 0.329	1.33 at 2026-06-03 21:57Z; dt 2.4 min
openweather	pressure	hPa	4 / 288	1012 to 1017; mean 1014.65	1015 at 2026-06-03 21:57Z; dt 2.4 min
openweather	temperature	degC	4 / 288	22.41 to 33.69; mean 26.1529	25.34 at 2026-06-03 21:57Z; dt 2.4 min
openweather	wind_direction	degree	4 / 288	0 to 360; mean 105.038	107 at 2026-06-03 21:57Z; dt 2.4 min
openweather	wind_speed	m/s	4 / 288	0 to 6.28; mean 2.1931	5.26 at 2026-06-03 21:57Z; dt 2.4 min
purpleair	pm25	ug/m3	61 / 4440	0 to 40.4; mean 6.7204	5 at 2026-06-03 22:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:32Z; dt 147.8 min
sentinel5p	s5p_ch4_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:32Z; dt 147.8 min
sentinel5p	s5p_co_column	mol/m^2	5 / 5	0.0277457 to 0.0294826; mean 0.0285	0.0294826 at 2026-06-03 19:04Z; dt 175.8 min
sentinel5p	s5p_co_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:32Z; dt 147.8 min
sentinel5p	s5p_hcho_column	mol/m^2	3 / 3	0.000135539 to 0.000169173; mean 0.0001	0.000135539 at 2026-06-04 18:45Z; dt 1245.3 min
sentinel5p	s5p_hcho_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:32Z; dt 147.8 min
sentinel5p	s5p_no2_column	mol/m^2	4 / 4	4.04755e-05 to 4.7738e-05; mean 0	4.64928e-05 at 2026-06-04 18:45Z; dt 1245.3 min
sentinel5p	s5p_no2_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:32Z; dt 147.8 min
sentinel5p	s5p_o3_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:32Z; dt 147.8 min
sentinel5p	s5p_so2_column	mol/m^2	4 / 4	-9.61344e-06 to 1.55388e-06; mean -0	1.55388e-06 at 2026-06-04 18:45Z; dt 1245.3 min
sentinel5p	s5p_so2_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:32Z; dt 147.8 min
tceq	co	ppm	3 / 217	0 to 0.7; mean 0.1774	0.2 at 2026-06-03 22:00Z; dt 0 min
tceq	no2	ppb	12 / 836	-2.6 to 28.2; mean 7.7523	22.4 at 2026-06-03 22:00Z; dt 0 min
tceq	so2	ppb	3 / 215	-0.6 to 3.6; mean -0.0056	-0.5 at 2026-06-03 22:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-02 06Z 94.98, 12Z 75.84, 18Z 63.1, 06-03 00Z 82.43, 06Z 88.58, 12Z 80.04, 18Z 90.07, 06-04 00Z 92.96, 06Z 95.66, 12Z 87.01, 18Z 75.78, 06-05 00Z 86.18, 06Z 91.84
asos	temperature	06-02 06Z 24.81, 12Z 28.81, 18Z 30.72, 06-03 00Z 25.09, 06Z 24.03, 12Z 26.43, 18Z 24.7, 06-04 00Z 23.92, 06Z 23.48, 12Z 25.68, 18Z 28.06, 06-05 00Z 26.05, 06Z 25.13
asos	wind_direction	06-02 06Z 15.38, 12Z 96.19, 18Z 106.6, 06-03 00Z 80, 06Z 153.2, 12Z 83.41, 18Z 79.51, 06-04 00Z 70, 06Z 63.75, 12Z 62.62, 18Z 114.8, 06-05 00Z 104.6, 06Z 106.8
asos	wind_speed	06-02 06Z 0.1187, 12Z 2.119, 18Z 4.291, 06-03 00Z 4.468, 06Z 3.1, 12Z 3.442, 18Z 2.848, 06-04 00Z 2.695, 06Z 1.916, 12Z 2.364, 18Z 5.242, 06-05 00Z 3.074, 06Z 3.509
noaa_gfs	gh_500	06-02 12Z 5881, 18Z 5894, 06-03 00Z 5880, 06Z 5881, 12Z 5869, 18Z 5878, 06-04 00Z 5868, 06Z 5875, 12Z 5870, 18Z 5879, 06-05 00Z 5876, 06Z 5884
noaa_gfs	pbl_height	06-02 12Z 45.64, 18Z 1521, 06-03 00Z 727.2, 06Z 223, 12Z 191.5, 18Z 610.5, 06-04 00Z 371.9, 06Z 149.4, 12Z 112.5, 18Z 1061, 06-05 00Z 695.2, 06Z 395.4
noaa_gfs	precipitable_water	06-02 12Z 50.87, 18Z 49.97, 06-03 00Z 47.9, 06Z 49.95, 12Z 52.68, 18Z 53.52, 06-04 00Z 51.48, 06Z 51.05, 12Z 49.73, 18Z 52.47, 06-05 00Z 54.89, 06Z 52.46
noaa_gfs	surface_pressure	06-02 12Z 1012, 18Z 1012, 06-03 00Z 1011, 06Z 1013, 12Z 1014, 18Z 1014, 06-04 00Z 1012, 06Z 1013, 12Z 1013, 18Z 1013, 06-05 00Z 1011, 06Z 1011
noaa_gfs	t_850	06-02 12Z 17.49, 18Z 16.94, 06-03 00Z 19.21, 06Z 18.92, 12Z 16.68, 18Z 16.36, 06-04 00Z 16.85, 06Z 16.76, 12Z 16.53, 18Z 16.57, 06-05 00Z 17.06, 06Z 17.2
noaa_gfs	u_10m	06-02 12Z -0.308, 18Z -1.999, 06-03 00Z -2.732, 06Z -2.664, 12Z -1.286, 18Z -3.464, 06-04 00Z -2.667, 06Z -2.037, 12Z -1.613, 18Z -3.449, 06-05 00Z -3.706, 06Z -3.12

Source	Metric	Values
noaa_gfs	v_10m	06-02 12Z -1.015, 18Z -1.143, 06-03 00Z 2.006, 06Z 0.7707, 12Z -2.43, 18Z -0.9612, 06-04 00Z 0.353, 06Z -0.2736, 12Z -0.5237, 18Z -0.1025, 06-05 00Z 2.332, 06Z 1.619
openaq	ozone	06-02 06Z 0.004818, 12Z 0.01031, 18Z 0.03372, 06-03 00Z 0.02268, 06Z 0.02027, 12Z 0.01989, 18Z 0.02069, 06-04 00Z 0.01219, 06Z 0.009681, 12Z 0.01084, 18Z 0.02447, 06-05 00Z 0.01993, 06Z 0.02267
openaq	pm10	06-02 06Z 23, 12Z 31.17, 18Z 28.42, 06-03 00Z 17.5, 06Z 15.83, 12Z 22.08, 18Z 15.27, 06-04 00Z 18.33, 06Z 20.67, 12Z 15.08, 18Z 13.17, 06-05 00Z 21.83, 06Z 20.9
openaq	pm25	06-02 06Z 5.446, 12Z 6.923, 18Z 5.397, 06-03 00Z 5.229, 06Z 5.087, 12Z 7.347, 18Z 5.526, 06-04 00Z 6.826, 06Z 8.505, 12Z 8.012, 18Z 4.202, 06-05 00Z 5.031, 06Z 5.373
openweather	cloud_cover	06-02 06Z 26.88, 12Z 58.67, 18Z 84.46, 06-03 00Z 32.5, 06Z 89.04, 12Z 97.54, 18Z 82.88, 06-04 00Z 97.29, 06Z 98.17, 12Z 68.42, 18Z 50.08, 06-05 00Z 40.29, 06Z 42.12
openweather	humidity	06-02 06Z 93, 12Z 78.75, 18Z 69.12, 06-03 00Z 84.5, 06Z 86.21, 12Z 83.04, 18Z 84.62, 06-04 00Z 92.12, 06Z 93.5, 12Z 87.92, 18Z 78.71, 06-05 00Z 85.75, 06Z 91.31
openweather	precipitation	06-02 06Z 0, 12Z 0, 18Z 0.8608, 06-03 00Z 0.1837, 06Z 0, 12Z 0.6279, 18Z 1.722, 06-04 00Z 0, 06Z 0.06292, 12Z 0.3583, 18Z 0, 06-05 00Z 0, 06Z 0.1975
openweather	pressure	06-02 06Z 1014, 12Z 1015, 18Z 1013, 06-03 00Z 1015, 06Z 1015, 12Z 1017, 18Z 1015, 06-04 00Z 1015, 06Z 1015, 12Z 1016, 18Z 1013, 06-05 00Z 1014, 06Z 1013
openweather	temperature	06-02 06Z 25.25, 12Z 29.24, 18Z 31.49, 06-03 00Z 24.58, 06Z 23.82, 12Z 26.55, 18Z 25.81, 06-04 00Z 23.73, 06Z 23.18, 12Z 25.69, 18Z 28.57, 06-05 00Z 25.99, 06Z 25.15
openweather	wind_direction	06-02 06Z 0, 12Z 125.3, 18Z 140.6, 06-03 00Z 98.12, 06Z 82.88, 12Z 55.38, 18Z 141.1, 06-04 00Z 97.83, 06Z 123.3, 12Z 90.58, 18Z 123.2, 06-05 00Z 114.7, 06Z 101.1
openweather	wind_speed	06-02 06Z 0, 12Z 1.848, 18Z 2.454, 06-03 00Z 2.2, 06Z 1.488, 12Z 2.451, 18Z 4.09, 06-04 00Z 1.993, 06Z 1.326, 12Z 2.228, 18Z 2.691, 06-05 00Z 2.174, 06Z 2.061
purpleair	pm25	06-02 06Z 6.367, 12Z 6.804, 18Z 4.413, 06-03 00Z 5.961, 06Z 6.557, 12Z 8.375, 18Z 5.832, 06-04 00Z 8.448, 06Z 11.52, 12Z 8.678, 18Z 3.316, 06-05 00Z 5.273, 06Z 5.41
sentinel5p	s5p_aer_ai_granule_available	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	s5p_ch4_granule_available	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	s5p_co_column	06-02 18Z 0.02882, 06-03 18Z 0.02948, 06-04 18Z 0.02776
sentinel5p	s5p_co_granule_available	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	s5p_hcho_column	06-02 18Z 0.0001692, 06-04 18Z 0.0001378
sentinel5p	s5p_hcho_granule_available	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	s5p_no2_column	06-02 18Z 4.07e-05, 06-04 18Z 4.712e-05
sentinel5p	s5p_no2_granule_available	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	s5p_o3_granule_available	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	s5p_so2_column	06-02 18Z -8.155e-06, 06-04 18Z -4.03e-06
sentinel5p	s5p_so2_granule_available	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
tceq	co	06-02 06Z 0.3833, 12Z 0.2056, 18Z 0.1667, 06-03 00Z 0.1389, 06Z 0.1222, 12Z 0.1667, 18Z 0.2, 06-04 00Z 0.1611, 06Z 0.1444, 12Z 0.1882, 18Z 0.1889, 06-05 00Z 0.2333, 06Z 0.14
tceq	no2	06-02 06Z 7.375, 12Z 7.831, 18Z 6.415, 06-03 00Z 6.2, 06Z 6.664, 12Z 9.104, 18Z 13.05, 06-04 00Z 9.914, 06Z 8.826, 12Z 9.671, 18Z 4.491, 06-05 00Z 5.191, 06Z 4.828
tceq	so2	06-02 06Z -0.1, 12Z -0.01176, 18Z 0.1556, 06-03 00Z -0.01667, 06Z -0.02778, 12Z 0.55, 18Z 0.1313, 06-04 00Z -0.2056, 06Z -0.2556, 12Z -0.1529, 18Z -0.02778, 06-05 00Z -0.06667, 06Z -0.12

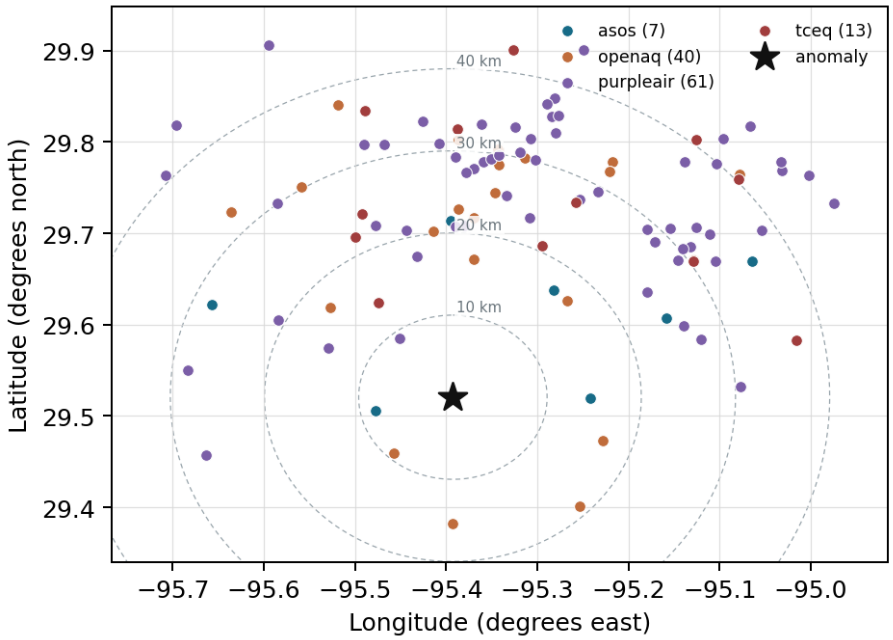
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

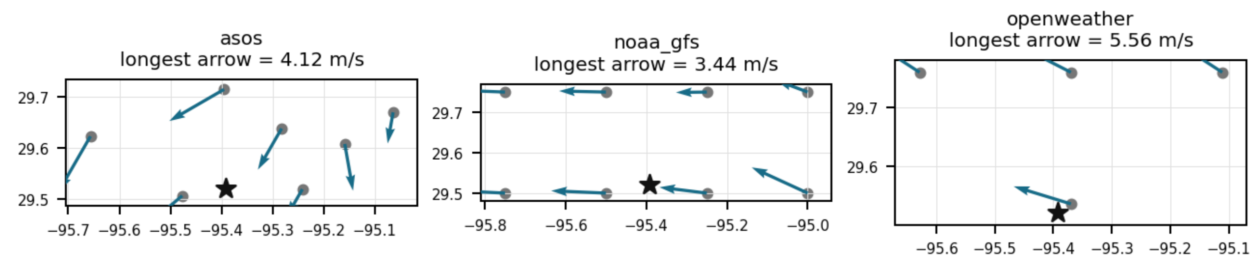
Source	Metric	Values
asos	humidity	AXH (8.32 km) 78.55, LVJ (14.594 km) 84.54, HOU (16.816 km) 82.66, MCJ (21.523 km) 85.62, EFD (24.589 km) 87.35, SGR (27.929 km) 84.57, T41 (35.803 km) 86.33
asos	temperature	AXH (8.32 km) 26.36, LVJ (14.594 km) 26.01, HOU (16.816 km) 26.19, MCJ (21.523 km) 25.63, EFD (24.589 km) 25.95, SGR (27.929 km) 26.13, T41 (35.803 km) 25.9
asos	wind_direction	AXH (8.32 km) 76.39, LVJ (14.594 km) 92.08, HOU (16.816 km) 92.78, MCJ (21.523 km) 102.3, EFD (24.589 km) 88.46, SGR (27.929 km) 101.2, T41 (35.803 km) 85.28
asos	wind_speed	AXH (8.32 km) 1.936, LVJ (14.594 km) 2.529, HOU (16.816 km) 3.544, MCJ (21.523 km) 4.253, EFD (24.589 km) 3.245, SGR (27.929 km) 3.118, T41 (35.803 km) 3.465
noaa_gfs	gh_500	gfs:29.50,-95.50 (10.647 km) 5878, gfs:29.50,-95.25 (13.977 km) 5878, gfs:29.75,-95.50 (27.558 km) 5878, gfs:29.75,-95.25 (29.004 km) 5878, gfs:29.50,-95.75 (34.669 km) 5879, gfs:29.50,-95.00 (38.051 km) 5878, gfs:29.75,-95.75 (42.957 km) 5878, gfs:29.75,-95.00 (45.723 km) 5877
noaa_gfs	pbl_height	gfs:29.50,-95.50 (10.647 km) 471.7, gfs:29.50,-95.25 (13.977 km) 451.6, gfs:29.75,-95.50 (27.558 km) 625, gfs:29.75,-95.25 (29.004 km) 548.5, gfs:29.50,-95.75 (34.669 km) 506.3, gfs:29.50,-95.00 (38.051 km) 507.6, gfs:29.75,-95.75 (42.957 km) 524.7, gfs:29.75,-95.00 (45.723 km) 434
noaa_gfs	precipitable_water	gfs:29.50,-95.50 (10.647 km) 51.4, gfs:29.50,-95.25 (13.977 km) 52.14, gfs:29.75,-95.50 (27.558 km) 50.55, gfs:29.75,-95.25 (29.004 km) 51.58, gfs:29.50,-95.75 (34.669 km) 50.55, gfs:29.50,-95.00 (38.051 km) 52.59, gfs:29.75,-95.75 (42.957 km) 49.95, gfs:29.75,-95.00 (45.723 km) 52.55
noaa_gfs	surface_pressure	gfs:29.50,-95.50 (10.647 km) 1012, gfs:29.50,-95.25 (13.977 km) 1013, gfs:29.75,-95.50 (27.558 km) 1012, gfs:29.75,-95.25 (29.004 km) 1013, gfs:29.50,-95.75 (34.669 km) 1011, gfs:29.50,-95.00 (38.051 km) 1014, gfs:29.75,-95.75 (42.957 km) 1010, gfs:29.75,-95.00 (45.723 km) 1014
noaa_gfs	t_850	gfs:29.50,-95.50 (10.647 km) 17.24, gfs:29.50,-95.25 (13.977 km) 17.16, gfs:29.75,-95.50 (27.558 km) 17.31, gfs:29.75,-95.25 (29.004 km) 17.23, gfs:29.50,-95.75 (34.669 km) 17.31, gfs:29.50,-95.00 (38.051 km) 17.11, gfs:29.75,-95.75 (42.957 km) 17.23, gfs:29.75,-95.00 (45.723 km) 17.12
noaa_gfs	u_10m	gfs:29.50,-95.50 (10.647 km) -2.401, gfs:29.50,-95.25 (13.977 km) -2.397, gfs:29.75,-95.50 (27.558 km) -2.166, gfs:29.75,-95.25 (29.004 km) -2.309, gfs:29.50,-95.75 (34.669 km) -2.358, gfs:29.50,-95.00 (38.051 km) -3.259, gfs:29.75,-95.75 (42.957 km) -2.048, gfs:29.75,-95.00 (45.723 km) -2.424
noaa_gfs	v_10m	gfs:29.50,-95.50 (10.647 km) -0.2576, gfs:29.50,-95.25 (13.977 km) -0.07681, gfs:29.75,-95.50 (27.558 km) -0.08348, gfs:29.75,-95.25 (29.004 km) -0.1001, gfs:29.50,-95.75 (34.669 km) -0.1985, gfs:29.50,-95.00 (38.051 km) 0.8365, gfs:29.75,-95.75 (42.957 km) -0.05764, gfs:29.75,-95.00 (45.723 km) 0.359
openaq	ozone	312 (13.954 km) 0.01628, 320 (16.839 km) 0.01773, 325 (20.732 km) 0.01674, 311 (22.058 km) 0.02008, 3378 (27.085 km) 0.01662, 332 (30.466 km) 0.02278, 2039 (32.161 km) 0.01486, 287 (32.585 km) 0.02296, +4 more
openaq	pm10	13380082 (27.085 km) 21.96, 271 (30.466 km) 18.32
openaq	pm25	14152710 (9.275 km) 7.945, 11987924 (15.421 km) 1.905, 11638043 (16.711 km) 2.114, 9416627 (16.97 km) 4.441, 15399615 (16.982 km) 5.39, 16026701 (18.852 km) 1.321, 8967001 (20.364 km) 5.609, 16123962 (21.937 km) 5.104, +18 more
openweather	cloud_cover	grid:south (2.782 km) 72.99, grid:center (26.771 km) 68.86, grid:west (35.1 km) 66.86, grid:east (38.099 km) 70.08
openweather	humidity	grid:south (2.782 km) 82.72, grid:center (26.771 km) 83.68, grid:west (35.1 km) 83.39, grid:east (38.099 km) 88.92
openweather	precipitation	grid:south (2.782 km) 0.265, grid:center (26.771 km) 0.3279, grid:west (35.1 km) 0.2929, grid:east (38.099 km) 0.43
openweather	pressure	grid:south (2.782 km) 1015, grid:center (26.771 km) 1015, grid:west (35.1 km) 1015, grid:east (38.099 km) 1015
openweather	temperature	grid:south (2.782 km) 26.26, grid:center (26.771 km) 26.17, grid:west (35.1 km) 26.12, grid:east (38.099 km) 26.06
openweather	wind_direction	grid:south (2.782 km) 102.2, grid:center (26.771 km) 101.9, grid:west (35.1 km) 123.6, grid:east (38.099 km) 92.44
openweather	wind_speed	grid:south (2.782 km) 2.195, grid:center (26.771 km) 1.963, grid:west (35.1 km) 1.534, grid:east (38.099 km) 3.079

Source	Metric	Values
purpleair	pm25	161003 (9.171 km) 4.255, 161015 (14.484 km) 8.445, 166451 (17.611 km) 8.693, 293735 (20.739 km) 9.255, 276512 (20.79 km) 7.605, 46237 (20.878 km) 0.174, 293725 (20.905 km) 9.392, 186315 (22.472 km) 8.845, +53 more
tceq	co	482011035 (27.064 km) 0.1667, 482011039 (30.447 km) 0.1288, 482011052 (32.687 km) 0.2375
tceq	no2	480391004 (0.0 km) 5.849, 482010416 (20.723 km) 10.59, 482010055 (22.053 km) 8.197, 482011066 (24.373 km) 11.42, 482011035 (27.064 km) 8.34, 482011039 (30.447 km) 7.615, 482011052 (32.687 km) 9.672, 482010047 (36.114 km) 10.6, +4 more
tceq	so2	482010051 (13.97 km) -0.2029, 482011035 (27.064 km) -0.09863, 482011039 (30.447 km) 0.274

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 35

A localized fresh-combustion plume is supported because the anomalous TCEQ monitor measured 22.4 ppb NO2 at 2026-06-03T22:00Z while the same site had a period mean of 5.849 ppb and the network 6-hour mean was 13.05 ppb, indicating a site-enhanced spike rather than only a region-wide elevation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 35

Houston meteorological conditions near 2026-06-03T22:00:00+00:00 favored short-term NO2 persistence because humidity and cloud cover were high, light rain was present, surface winds were only modest, and forecast planetary boundary layer height was relatively low to moderate.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 35

A nearby fresh combustion emission plume from urban traffic, diesel activity, marine or rail operations, or industrial combustion is the most plausible direct cause of the 22.4 ppb surface NO2 spike at the Houston TCEQ monitor because NO2 was sharply elevated at the site while ozone remained low and particulate matter remained unremarkable, which is the signature of freshly emitted NOx rather than aged regional pollution.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 35

NOAA GFS model output shows the regional planetary boundary layer height collapsing from a 6-hour mean of 610.5 meters at 18:00 UTC to 371.9 meters by 00:00 UTC on June 4, severely limiting vertical pollutant dispersion.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 35

The air quality anomaly in Houston, Texas on June 3rd was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 35

The anomaly occurred on June 3rd at approximately 21:57:35 UTC, with the nearest sensor located 9.171 km away.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 35

The regional 6-hour average NO_2 concentration across TCEQ monitoring stations peaked at 13.05 ppb during the 18:00 UTC to 00:00 UTC window on June 3, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 35

NO_2 levels were also elevated, with readings up to 28.2 ppb recorded by TCEQ monitors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 35

A strong wind pattern may have blown pollutants from nearby industrial sites or traffic congestion into the affected area, causing the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 35

Dense cloud cover exceeding 80 percent and light easterly surface winds near 3 m/s reduced photolysis rates and facilitated local NO_2 accumulation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 35

Meteorological support for short-term NO2 persistence is present because conditions near the anomaly were humid and cloudy with ongoing light rain and only modest boundary-layer dilution, including about 84.6 to 89 percent humidity, about 95 percent cloud cover, about 1.33 mm/h precipitation, and GFS planetary boundary layer height near 324 m around the event window.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 35

Extensive cloud cover and ongoing precipitation suppressed solar irradiance, reducing NO2 photolysis rates and allowing NO2 to accumulate.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 35

The temperature was unusually low on June 3rd, which may have contributed to the formation of ground-level ozone.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 35

The air quality anomaly in Houston, Texas is caused by a combination of high temperatures and wind direction.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 35

The TCEQ NO2 measurement at location 29.520, -95.393 was 22.4 ppb at 2026-06-03T22:00:00+00:00, while the expected value was 3.6930232558139537 ppb and the reported z-score was 6.33457703104176.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 35

The air quality was elevated above the mean value of 6.7204 ug/m3 for PM2.5 in the surrounding area, with a range of 0.0-40.4 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 35

The Houston air-quality pattern at 2026-06-03T22:00:00+00:00 is most consistent with a localized fresh-combustion NOx plume rather than a broad regional pollution event because ozone was low near the anomaly, PM2.5 and PM10 were not elevated, CO was only modest, and satellite CO and SO2 did not indicate a strong coincident regional plume.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 35

TCEQ ground monitoring data documented an NO2 concentration spike to 22.4 ppb at 22:00 UTC on June 3, 2026, during a period when the regional 6-hour average NO2 reached 13.05 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 35

A broad regional combustion or sulfur-rich industrial event is contradicted because co-located or nearby indicators did not show corresponding enhancements, with ozone near 0.014 ppm, PM2.5 near 5.0 to 7.4 ug/m3, CO near 0.2 ppm, SO2 near -0.5 ppb, and Sentinel-5P CO columns only modestly elevated rather than extreme.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 35

The air quality anomaly is related to particulate matter (PM2.5) with a value of 5.0 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 35

Planetary boundary layer height decreased to under 400 meters during the late afternoon of June 3, 2026, trapping surface pollutants.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 35

The pollutant identified as anomalous was nitrogen dioxide (NO_2) at a TCEQ monitor in Houston.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 35

A severe nitrogen dioxide anomaly was detected at coordinates (29.520, -95.393) on June 3, 2026, at 22:00:00 UTC, where the measured NO_2 concentration reached 22.4 ppb against an expected baseline of 3.69 ppb (z-score 6.33).

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 35

The air quality anomaly in Houston, Texas is not caused by CO levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 35

The TCEQ monitoring network detected a severe NO2 concentration anomaly reaching 22.4 ppb at 22:00 UTC on June 3, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 35

The TCEQ monitor at 29.520, -95.393 recorded a severe NO2 anomaly of 22.4 ppb at 2026-06-03T22:00:00+00:00, far above the site-specific expected value of 3.693 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 35

ASOS surface observations and OpenWeather data recorded cloud cover exceeding 80 percent alongside light easterly winds averaging approximately 3 m/s, which attenuated photochemical NO_2 loss while advecting regional emissions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 35

High levels of particulate matter ($PM_{2.5}$) were detected in the air, potentially exacerbating respiratory issues and contributing to the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 35

Boundary-layer dilution was limited enough to allow a local NO_2 plume to remain concentrated near the surface at the time of the anomaly because Houston-area winds were only light to moderate and GFS planetary boundary layer height near the event was a few hundred meters rather than deeply mixed daytime values above about 1 km.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 35

Cloudy, humid, lightly rainy conditions likely suppressed photochemical conversion and promoted heterogeneous retention of fresh NO_2 near the source region, making a transient surface NO_2 enhancement more likely to persist than under hot sunny afternoon conditions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 35

The PM2.5 levels were significantly higher than normal, with readings up to 40.4 ug/m3 recorded by PurpleAir sensors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 35

The accumulation of late-afternoon mobile source emissions during peak commuting hours combined with a shallowing planetary boundary layer trapped elevated NO_2 concentrations near the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 35

The anomalous TCEQ monitor was the monitor nearest the anomaly location, and its period mean NO2 concentration was 5.849 ppb compared with a 12-monitor TCEQ NO2 network mean of 7.7523 ppb over the context window.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 35

Low-speed easterly to east-northeasterly surface winds adducted local combustion plumes from upwind industrial and urban sources directly over the TCEQ monitoring site.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.